

HDOC – Day 26

C Programming Assignment

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Topic: Right-aligned patterns using switch statement and nested loops (without arrays)

Problem: Display a menu, accept the user choice, and print the selected right-aligned pattern. Each pattern is implemented using a separate parameterless function with no return value.

Algorithm

1. Display the menu and read choice.
2. According to the choice, call the corresponding parameterless function.
3. Read n and reject $n \leq 0$.
4. For each missing pattern position, print FOUR spaces.
5. For each actual star, print "*" .
6. For the number pattern, print the natural number followed by three spaces.
7. Print a newline after every row.
8. Stop.

C Program

```
#include <stdio.h>
```

```
void increasingStar(void);
```

```
void increasingNumber(void);
```

```
void decreasingStar(void);
```

```
int main(void)

{
int choice;

printf("RIGHT-ALIGNED PATTERNS MENU\n");
printf("1. Right-aligned increasing triangle\n");
printf("2. Right-aligned increasing number triangle\n");
printf("3. Right-aligned decreasing triangle\n");
printf("Enter your choice: ");

scanf("%d", &choice);

switch (choice)
{
case 1:
increasingStar();

break;

case 2:
increasingNumber();

break;

case 3:
decreasingStar();

break;

default:
printf("Invalid choice.\n");
}

return 0;

}
```

```

void increasingStar(void)
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    if (n <= 0) {
        printf("Invalid input. Enter a positive integer.\n");
        return;
    }
    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n - i; j++)
            printf(" "); /* four leading spaces */
        for (j = 1; j <= i; j++)
            printf("* "); /* star + three spaces */
        printf("\n");
    }
}

void increasingNumber(void)
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    if (n <= 0) {
        printf("Invalid input. Enter a positive integer.\n");
    }
}

```

```

return;

}

for (i = 1; i <= n; i++)
{
    for (j = 1; j <= n - i; j++)
        printf(" "); /* four leading spaces */
    for (j = 1; j <= i; j++)
        printf("%d ", j); /* number + three spaces */
    printf("\n");
}
}

void decreasingStar(void)
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);
    if (n <= 0) {
        printf("Invalid input. Enter a positive integer.\n");
        return;
    }
    for (i = n; i >= 1; i--)
    {
        for (j = 1; j <= n - i; j++)
            printf(" "); /* four leading spaces */
        for (j = 1; j <= i; j++)

```

```

printf("* "); /* star + three spaces */

printf("\n");

}

}

```

Test Case Table

TC	Choice	n	Expected output	Result
1	1	4	Right-aligned increasing star triangle with 4 rows	Pass
2	2	4	Rows: 1; 1 2; 1 2 3; 1 2 3 4, right-aligned	Pass
3	3	4	Right-aligned decreasing star triangle: 4,3,2,1 stars	Pass
4	4	-	Invalid choice.	Pass
5	1	0	Invalid number of rows.	Pass

Compilation, Execution & Testing

```

kali@kali: ~
File Actions Edit View Help
(kali@kali)-[~]
$ cc hd26.c
(kali@kali)-[~]
$ ./a.out
RIGHT-ALIGNED PATTERNS MENU
1. Right-aligned increasing star triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing star triangle
Enter your choice: 1
Enter number of rows: 4
      *
     * *
    * * *
   * * * *
(kali@kali)-[~]
$

```

```
kali@kali: ~  
File Actions Edit View Help  
(kali@kali)-[~]  
$ cc hd26.c  
(kali@kali)-[~]  
$ ./a.out  
RIGHT-ALIGNED PATTERNS MENU  
1. Right-aligned increasing star triangle  
2. Right-aligned increasing number triangle  
3. Right-aligned decreasing star triangle  
Enter your choice: 2  
Enter number of rows: 4  
    1  
  1 2  
1 2 3  
1 2 3 4  
(kali@kali)-[~]  
$
```

```
kali@kali: ~  
File Actions Edit View Help  
(kali@kali)-[~]  
$ cc hd26.c  
(kali@kali)-[~]  
$ ./a.out  
RIGHT-ALIGNED PATTERNS MENU  
1. Right-aligned increasing star triangle  
2. Right-aligned increasing number triangle  
3. Right-aligned decreasing star triangle  
Enter your choice: 3  
Enter number of rows: 4  
  * * * *  
   * * *  
    * *  
     *
```

```
kali@kali: ~  
File Actions Edit View Help  
(kali@kali)-[~]  
$ cc hd26.c  
(kali@kali)-[~]  
$ ./a.out  
RIGHT-ALIGNED PATTERNS MENU  
1. Right-aligned increasing star triangle  
2. Right-aligned increasing number triangle  
3. Right-aligned decreasing star triangle  
Enter your choice: 4  
Invalid choice.  
(kali@kali)-[~]  
$
```

```
kali@kali: ~  
File Actions Edit View Help  
(kali@kali)-[~]  
$ cc hd26.c  
(kali@kali)-[~]  
$ ./a.out  
RIGHT-ALIGNED PATTERNS MENU  
1. Right-aligned increasing star triangle  
2. Right-aligned increasing number triangle  
3. Right-aligned decreasing star triangle  
Enter your choice: 1  
Enter number of rows: 0  
Invalid number of rows.  
(kali@kali)-[~]  
$
```